Module 3: Lexical Semantics & Word Embeddings

WordNet, Lesk WSD, TF-IDF, Word2Vec (CBOW/SGNS), GloVe, FastText

Module III: Lexical Semantics & Word Embeddings — Topics Covered

1. WordNet Synsets & Lesk Disambiguation
2. TF-IDF Vector Space & Cosine Similarity
3. Word2Vec: Continuous Bag of Words (CBOW)
4. Word2Vec: Skip-Gram with Negative Sampling
5. GloVe Global Vectors & FastText Subwords

1. Word2Vec Architectures

Architecture	Objective Formulation	Key Strength
Continuous Bag-of-Words (CBOW)	Predicts target center word w_t given sum of surrounding context words $w_{t-c}, \dots, w_{t+c}$.	Faster to train; better representation for frequent words.
Skip-Gram with Negative Sampling (SGNS)	Predicts surrounding context words given target center word w_t: $\max \sum \log \sigma(v'_c \cdot v_w) + \sum \mathbb{E}[\log \sigma(-v'_k \cdot v_w)]$.	Superior performance on rare words and fine-grained analogies.